RAJALAKSHMI ENGINEERING COLLEGE

Tharun Periyan A P 251001289-2025-IT

T2

Started on	Friday, 11 September 2026, 1:56 PM
State	Finished
Completed on	Friday, 11 September 2026, 2:10 PM
Time taken	13 mins 49 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int f(int a[],int l,int r,int x){
    if (l>r)
        return -1;
    int m=(l+r)/2;
    if (a[m]==x)
        return a[m];
    if (a[m]>x)
        return f(a,l,m-1,x);
    int faa=f(a,m+1,r,x);
    if (faa!=-1)
        return faa;
    return a[m];
}

int main(){
    int n,x;
    scanf("%d",&n);
    int a[n];
    for (int i=0;i<n;i++)
        scanf("%d",&a[i]);
    scanf("%d",&x);
    printf("%d",f(a,0,n-1,x));
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

	Input	Expected	Got	
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course